√Ma+YC

Virginia Mathematical Association of Two Year Colleges

Statewide Conference

Mathematics for Every Learner: Equity and Access

March 21 - 22, 2025

Dear colleagues,

Greetings and a warm welcome to the VMATYC spring meeting being held in the northern region! We are thankful to Piedmont Virginia Community College for hosting. We extend a special note of appreciation to Wendi Dass, Keith Nabb and Support Staff at Piedmont Virginia Community College, whose hard work and dedication have been instrumental in organizing and setting up the conference, and to the esteemed speakers who will be presenting their insightful talks.

Additionally, we are grateful for the breakfast provided by Lumen Learning and lunch provided by Pearson.

Please keep in mind the following announcements:

- Need a Campus Map? Check out the QR Code to the right.
- Consider joining VMATYC as a member! Join now to receive membership through December 2025. Visit http://vmatyc.org/join-vmatyc/
- To stay updated with the most recent VMATYC developments, make sure to visit our website at https://vmatyc.org/ and read our newsletter.

Campus Map

Matthew Westerhoff
Northern Virginia CC
VMATYC President - Elect

Michael Filsinger
Virginia Peninsula CC
VMATYC President

Agenda

Friday

Opening: Registration [South Entrance]

1:00 pm – 2:00 pm

Session One: 2:00 pm – 2:50 pm

Session Two: 3:00 pm – 3:50 pm

Session Three: 4:00 pm – 4:50 pm

Dinner Banquet: Keynote Speaker [Main Event Space]

5:30 pm – 6:30 pm

Saturday

Breakfast Catered by Lumen Learning [North Mall]

Session Four: 9:00 am – 9:50 am

Session Five: 10:00 am – 10:50 am

Session Six: 11:00 am – 11:50 am

Lunch Catered by Pearson [North Mall]

12:00 pm – 1:00 pm

Presentation Descriptions

Welcome Speaker: Dr. Jean Runyon, President of PVCC

Keynote Speaker: Casedy Thomas

Session One (2:00 pm – 2:50 pm)

Making Math Fun: Getting Students Involved with Hands-On Learning [M261] *Kerri Bentjen, Kinga Oliver and Robert Chaney,* Sinclair Community College

Discover Sinclair College innovative NSF-funded math curriculum, where foundational courses like College Algebra, Trigonometry, and Calculus come to life through interactive, real-world applications. Explore how robots and active learning techniques empower students to connect mathematical concepts to practical problem-solving. Join us for engaging demonstrations, hands-on activities, and insights from student feedback on this transformative approach.

Powerful Tools that Empower You and Your Students [M247] *Amy Horner, Maxim Young, Natalie Chakrain,* Lumen Learning

Attend this interactive workshop to discuss (share and/or learn) tools that Lumen Learning has to empower faculty and students.

Determine Area and Centroid of NACA Airfoil [M251] *Triet Truong, and Javier Torrez,* Northern Virginia Community College (Student Presentation)

The airfoil is a unique kind of curved structure that utilizes Bernoulli's Principle to create lift or drag which aid and allow airplanes to fly. Research was done on 78 different types of airfoils to test how different thicknesses, maximum chambers, and chord lengths affect the effectiveness of wings in airplanes. Each airfoil was assigned a NACA (National Advisory Committee for Aeronautics) number to tailor the general functions of an airfoil into the corresponding equations of the distinct NACA airfoil. In this presentation, we will discuss the approach for determining the area and centroid of the sub-region bounded by the airfoil structure.

Session Two (3:00 pm – 3:50 pm)

Pearson Education Innovations: AI-Powered Tool for Students & Instructors! [M261] *Catherine Hares,* Pearson Education

In this session, Pearson Education will focus on our recent innovations, specifically with the use of AI and how this is shaping the future of education. Discover how our AI Bot empowers you to create personalized learning for individual student needs, effortlessly create assignments, and transform your workflow. Learn how this tool is positively benefiting students with improved engagement, interactive content, and convenience. AI at Pearson helps us work smarter to accomplish your tasks and allows you to spend more time connecting with your students. As educators, we understand your challenges, and our AI tools are here to help. Join us to see how these innovations can make a difference to your students and in your teaching!

Collatz Conjecture [M251] *Chamath Hettiarachchi,* Northern Virginia Community College Annandale Campus

The behavior of a Collatz sequence is clearly related to the way in which the powers of 2 are distributed among the powers of 3. We were surprised to find that the powers of 2 appear to be bounded away from the powers of 3 by an amount which grows almost as rapidly as the power of 3. We will also identify those positive integers with a given coefficient stopping time.

Measuring the Impact of Corequisite Mathematics on Student Success [M259] *Zack Beamer,* Piedmont Virginia Community College

Over the past several years the VCCS has moved towards corequisite remediation for marginally prepared incoming college students. Across all 23 colleges, recent high school students with a GPA between 2.0 and 3.0 enroll in college-level courses with concurrent developmental learning support. Those below the 2.0 threshold begin in prerequisite coursework and those above the 3.0 threshold start directly into unsupported college-level mathematics. With all colleges participating in this model, we are now able to begin analyzing the impacts of placing students into corequisite (MDE) courses. With support from an NSF grant, a team of researchers including VCCS representatives is now exploring the impacts of corequisites on student success in mathematics.

Using a mixed-methods approach, this ongoing research study seeks to understand how placement into corequisite courses affects enrollment and completion in college level mathematics. This presentation will highlight some of the comparisons between students just above the GPA thresholds for gateway mathematics enrollment and completion. In brief, students placed into corequisite support did not perform better than their directly enrolled peers, and the placement had a negative impact on enrollment for some students. Our subsequent quantitative analysis explored reasons for why these corequisite courses did not have the desired positive impact for students near the GPA thresholds.

Session Three (4:00 pm – 4:50 pm)

XYZ Homework: Supporting OER with the Latest Features and Courses from Developmental Math to Calculus and Beyond [M247] *Daniel Breuer,* XYZ Homework

You don't have to choose between student affordability and instructional quality. XYZ is the best of both worlds! Come see how the XYZ Homework team can support your 1st- & 2nd-year college math courses with print, custom eBooks, custom assignments, and more. Ask us about new features that help with large course management!

MIDPOINT-EFFICIENT SDF INTEGRATION PRESENTATION [M251] *Diego Fonseca,* Northern Virginia Community College (Student Presentation)

This presentation touches on a way to reduce iterations ahead of time by having two separate rays and checking for distance intersections between these two rays. Knowing how many iterations we need to compute ahead of time can help us reduce unnecessary computation. We show how for every distance increases greater than our limit. We can increase the magnitude of our end ray so that it is more likely for our ray to intersect distances. We determine the intersection by using the midpoint of our distance between both rays and seeing if it is less than our average signed distance. As of the current implementation, we have decreased our time by a heavily varied 32% per frame. We conclude that this method could also be implemented towards various other raymarching optimizations, as long as it has a fixed start and end point.

Direct Enrollment Corequisite Implementation - Findings from the VCCS [M259] *Zack Beamer,* Piedmont Virginia Community College

While every one of the 23 VCCS schools has transitioned to corequisite MDE developmental courses over the past few years, relatively few aspects of reform implementation were fixed by system policy. Instead, the bulk of implementation details were left to individual colleges to determine. Each college could choose which corequisite courses to offer and whether developmental-level and credit-level courses would be linked in SIS. The modality and scheduling of course sections were also left up to colleges, as well as course staffing and placement into higher-level courses.

This fall, institutional representatives from nearly all colleges completed a survey responding to questions about how their college has chosen to implement Direct Enrollment reforms. This presentation/workshop will overview descriptive statistics on implementation and share what aspects have been implemented in considerably different ways. These findings will also serve as a springboard to talking about policy directions and future areas for research

Session Four (9:00 am – 9:50 am)

The Concept of Infinite Series [M247] *Jiseong Park,* Northern Virginia Community College Alexandria Campus

All the Calculus teachers have experienced a certain array of difficulties in helping students overcome the confusion involving the concept of series, its partial sum and the sequence of its terms, the limiting behavior of the terms versus the convergence of the series itself. My presentation will point out how to resolve this problem, if not the most ideal way, but a rather radical disruption in the use of language. In this presentation, I suggest the prohibition of the word "infinite series".

Curious calculus questions: Is there a better coordinate system out there? [M251] *Keith Nabb and Jordan Higgins,* Piedmont Virginia Community College

Calculus students frequently ask questions whose answers require mathematics beyond Calculus. A few of these questions and answers will be revealed here. The highlight of this session will be revealing one student's work on an especially thorny question, "Is there a better coordinate system out there?"

Workshop: Engaging students with Content Literacy and Visual Aids [M259] *Sharon White,* Laurel Ridge Community College

Using these two tools, I engage students in active learning through classroom discussions on Mathematical Concepts. First tool content reading: I use several books with short engaging stories to facilitate learning in the classroom. These stories create discussions that engage the students' interest. Second Tool real-world applications: I create useful dashboards to discuss visual aids and how to read visual aids.

Session Five (10:00 am – 10:50 am)

Buffet of Mathematical Problems: Modulo Matrices to Counting Problems [M261] *Ryan Young, Jeff Ellis , and Keith Nabb,* Piedmont Virginia Community College

In an interactive session, we will invite participants to explore a diverse selection of mathematical challenges that test intuition, logic, and problem-solving skills. From uncovering the hidden complexities of modular arithmetic in matrices (and its applications in cryptography) to dissecting counting problems, we invite the audience to engage with mathematical ideas in unexpected ways.

Round Table Discussion: A discussion of corequisites [M251] *David French,* Tidewater Community College Chesapeake Campus

This will be a roundtable, open discussion of corequisites within the VCCS. This discussion will include different techniques, what schools and individuals are using, and experiences good and bad.

I am able to share some data from Tidewater, and will be willing to collect information from other institutions.

Effective Use of Artificial Intelligence: Google Notebook LM [M247] *Samuel Chukwuemeka,* Blue Ridge Community College, Weyers Cave

Can artificial intelligence (AI) be used effectively in teaching and learning?
Can we view the main points of a recorded meeting without listening to the entire recording? Attend this session to explore the valuable features of this AI tool: The Google Notebook Language Model (NotebookLM).

Encouraging Prepared Students to take Math [M259] *Alison Thimblin,* Northern Virginia Community College, Woodbridge

Students are expected (but unlikely) to complete college level math in their first year. Are some students ready for MTH 154, but placed into a corequisite pairing? How would we know? Would the opportunity to demonstrate readiness improve the number of students who take college level math? Let's explore some potential options.

Session Six (11:00 am – 11:50 am)

Roundtable Discussion: Acceleration and Hybridization in Mathematics Teaching [M261] *Keith Nabb and Wendi Dass*, Piedmont Virginia Community College

This session will bring together faculty to examine questions about shortening the academic calendar and placing some instructional materials online: What are the impacts of teaching mathematics in a 7-week format? What instruction is most effective? How does student learning change? Whether you are new or experienced with such innovation, please join us and contribute!

The Mathematics of Finance using TI Graphing Calculators [M247] *Samuel Chukwuemeka,* Blue Ridge Community College, Weyers Cave

The session will discuss the solutions of several problems in financial math topics using the Texas Instruments (TI) calculators. TI-84 Plus CE calculator shall be used.

Round Table Discussion: Math Placement past Gateway Courses - Strategies and Results [M251] *Lynne Ryan,* Blue Ridge Community College

When the math placement policy was changed to no longer require the VPT, placement past gateway courses (e.g. allowing students to place directly into Calculus w/o completion of VCCS MTH 161/162=167) was left to the individual colleges. As you may know, we were recently surveyed by the System Office as to what policies and procedures each college had put in place. This session would be an opportunity for us to discuss amongst faculty and compare notes. I would like to start with a brief presentation on the approach BRCC took and the three years of data we collected to measure its success, and then open the floor for a general discussion for how different colleges are handling this. If you have data from your college, you are encouraged to bring it!

Vendors for the Conference are set up in the tutoring center M253 in between the conference rooms.